

1 What is risk management?

Uncertainty that matters.

Example: $Q \times P$ (market PnL)

Decomposition

$$\begin{aligned} \text{pnl} &= Q_{t+1}P_{t+1} - Q_tP_t \\ &= (Q_{t+1}P_{t+1} - Q_tP_{t+1}) \\ &\quad + (Q_tP_{t+1} - Q_tP_t) \quad (\text{telescoping}) \\ &= \underbrace{(Q_{t+1} - Q_t)P_{t+1}}_{\text{trading_pnl}} + \underbrace{Q_t(P_{t+1} - P_t)}_{\text{market_pnl}} \end{aligned}$$

SK Hynix (000660.KS), KRW

	t	t+1
Q	100	120
P	1,919,000	1,783,000
Q · P	191.9M	214.0M
market_pnl = -13.6M		
trading_pnl = +35.7M		
Δ = +22.1M		

Trading Book | EQKOR
+ trading_pnl (stock bought)

→

Treasury Book | TRSYUS
- trading_pnl (cash lent)

Continue: market PnL

market_pnl = $Q_t \Delta P$. Observed price: P . Model price: $\hat{P}_t = f(r_t, S_t, \sigma_t)$.

$$\Delta P = \underbrace{(\Delta P - \Delta \hat{P})}_{\text{model risk / residual}} + \underbrace{\Delta \hat{P}}_{\text{modelled move}}$$

$$\begin{aligned} \Delta \hat{P} &= [f(r_{t+1}, S_t, \sigma_t) - f(r_t, S_t, \sigma_t)] \\ &\quad + [f(r_{t+1}, S_{t+1}, \sigma_t) - f(r_{t+1}, S_t, \sigma_t)] \\ &\quad + [f(r_{t+1}, S_{t+1}, \sigma_{t+1}) - f(r_{t+1}, S_{t+1}, \sigma_t)] \quad (\text{telescoping}) \end{aligned}$$

rate + stock + volatility = modelled market-PnL drivers.

- Price change is decomposed.
- Choose f so **model risk is minimized**.

2 Why statistics still matter in the age of AI

Statistics loop

Data → Statistical model → Metrics → Data

AI crux: formulate a loop.

Data → Agent → Feedback → Data

Looping for a good **model** to minimize model risk?

- **Data** = numbers + text
- **Agent** = LLM + tools
- **Feedback** = text (plain or tool call)

3 AI in Practice

3a. Data

3a.1 SQL data (DuckDB)

DuckDB – query databases and files with SQL.

```
pip install duckdb
```

Postgres via connection string

```
INSTALL postgres; LOAD postgres;  
ATTACH 'postgresql://user:password@host:5432/dbname'  
  AS pg (TYPE POSTGRES);  
SELECT * FROM pg.public.trades LIMIT 10;
```

S3 via httpfs

```
INSTALL httpfs; LOAD httpfs;  
SELECT * FROM read_parquet('s3://my-bucket/trades/*.parquet')  
LIMIT 10;
```

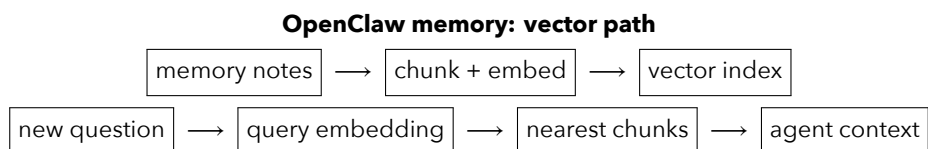
3a.2 Vector data

Vectors are transformed text. Qdrant would be a strong candidate to work on this data.

```
from sentence_transformers import SentenceTransformer  
v = SentenceTransformer("all-MiniLM-L6-v2").encode("risk management")  
# v: [0.12, -0.04, 0.09, ...] (384 numbers)
```

$$\text{cosine}(v, w) = \frac{v \cdot w}{\|v\| \|w\|} \quad \text{near } 1 = \text{similar}$$

``risk management`` vs. ``managing risk`` → high score.



Default OpenClaw search can also combine keyword matching; this shows the vector path.

3b. Agent runtime

Assistant UI demo – scaffold a standalone chat app with assistant-ui (React / Next.js chat UI).

Create a new standalone app

```
npx assistant-ui@latest create my-app  
cd my-app
```

Optional templates (pick one when scaffolding):

```
npx assistant-ui@latest create my-app --template minimal  
npx assistant-ui@latest create my-app --example with-ai-sdk-v7
```

Glossary. **LLM**: large language model; **PnL**: profit and loss; **S3**: object storage; **SQL**: Structured Query Language.